

# Krishnaa Vadivel

Scientific Computing | Machine Learning | Electrical Engineering

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Portfolio: <https://krishnaa423.github.io/personal-website/> | GitHub: <https://github.com/krishnaa423>

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## Summary

Electrical-engineering PhD researcher who bridges computational materials physics, GPU/HPC scientific software, scientific ML, and embedded engineering. Select this broad version only when a role genuinely crosses those domains.

## Technical Skills

**Scientific compute:** Python, C++, CUDA (cuBLAS, cuTENSOR), ROCm (rocBLAS, rocFFT), MPI, OpenMP, PETSc, SLEPc, DOLFINx, JAX, PyTorch Geometric

**Systems:** Frontier, Perlmutter, Docker, Kubernetes, Linux, CMake, ASP.NET, Microsoft SQL Server

**Hardware:** Verilog, Lattice FPGA, PIC32, dsPIC, STM32, C8051, board debugging, OpenGL

## Experience

### PhD Researcher

*Yale University, Electrical Engineering*

2021–present

- Translate many-body and first-principles physics into distributed Python/C++ workflows using MPI, OpenMP, PETSc/SLEPc, CUDA/ROCm, and national-lab GPU platforms.
- Build DeePH-style computational graphs and scientific-ML models with equivariant GNNs and automatic differentiation for LiF and double-perovskite optical-property calculations.
- Combine simulation research with semiconductor and photonics lab experience; publish and present self-trapped-exciton research.

### Software and Embedded Systems Engineer

*Probe Technologies / Weatherford Wireline Services, Houston, TX*

2019–2021

- Shipped CUDA/OpenCL and OpenGL applications for acoustic-waveform processing and diagnostic visualization.
- Delivered C#, ASP.NET, and SQL Server systems for remote engineering-tool testing and sensor-calibration data.
- Implemented firmware/FPGA functionality and supported board-level integration, debugging, and field validation.

## Selected Projects

**Materials ML** — Equivariant GNN and automatic-differentiation workflows for optical-property calculations.

**Integrated photonics** — Meep FDTD ring-resonator filter simulation, later fabricated and optically tested.

**Finite elements** — DOLFINx/PETSc heat, electrostatic, and flow simulations.

## Education

### PhD, Electrical Engineering

*Yale University*

2021–present

### MPhil and MS, Electrical Engineering

*Yale University*

2023

### MEng and BS, Electrical and Computer Engineering

*Cornell University,*

2013–2018